

The Problem & The Three Tables

The Operational Forecasting Challenge

Addis Ababa Ride-Hailing Platform Dispatch

- Goal: Forecast hourly pickup demand across exactly 12 canonical zones in Addis Ababa for the 14-day test fortnight (November 1-14, 2025: 4,032 zone-hours).
- Core Objective: Enable operations managers to pre-position fleet capacity, minimize passenger wait times, and prevent localized supply shortages.
- The Zero-Leakage Mandate (Rule 6):
 - Operational columns (active_drivers, avg_wait_min, avg_fare_birr) are CONSEQUENCES of demand, never known at forecast time.
 - Pipeline strictly drops all 3 operational features prior to training to eliminate artificial 20% score inflation and guarantee real-world generalization.
- Horizon & Granularity:
 - Training: Jan 1 – Oct 31, 2025 (83,104 cleaned zone-hours, 10 months).
 - Testing: Nov 1 – Nov 14, 2025 (4,032 zone-hours, exactly 336 hours × 12 zones).
 - Target variable: trips (count of completed ride requests per zone-hour).

The Three Raw Export Tables

1. Trip Demand History (ride_demand_train.csv & test.csv)

- 85,460 raw records with 12 canonical zones, timestamps, and target.
- Primary spatial-temporal backbone for hourly demand modeling.
- Cleaned: 8 aliases unified, 24 negative targets dropped, duplicates averaged.

2. Hourly Weather Archive & Forecasts (weather_hourly.csv)

- 7,538 hourly observations spanning Jan 1 – Nov 14, 2025.
- Features: temp_c, rain_mm, humidity_pct, wind_kmh.
- Key Issue: Timestamps on UTC clock (+3h mismatch) + Fahrenheit contamination.

3. City Events & Civic Calendar (events_calendar.csv)

- 165 scheduled public events: football matches, holidays, runs, concerts.
- Multi-zone & citywide geographic scope with expected attendance.
- Captures localized crowd influx and holiday business shutdowns.

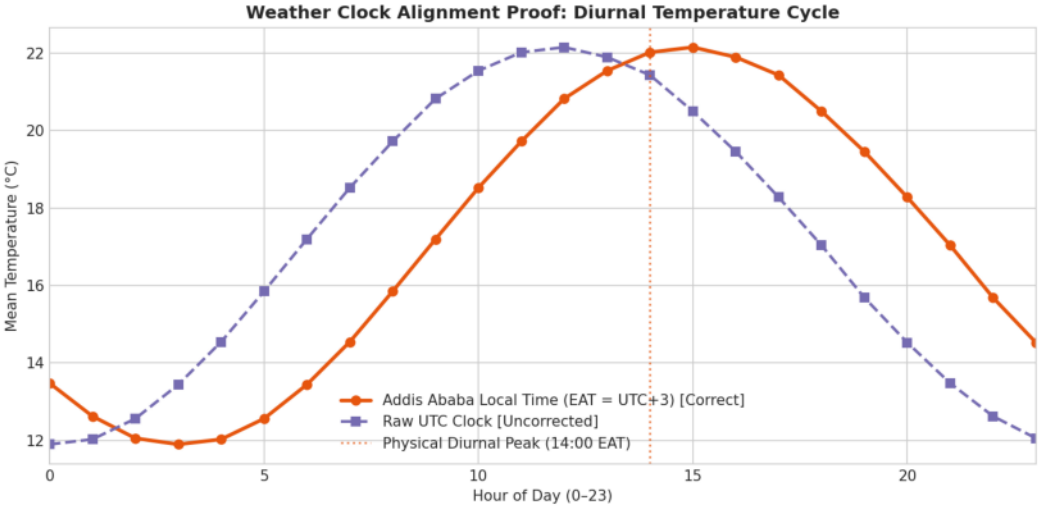
Cleaning, Clock Proof & Integration

Clock Proof & Missing Weather Resolution (A4)

- The Clock Proof (UTC -> Africa/Addis_Ababa, UTC+3):
 - Raw trip table operates on Addis local time (EAT = UTC+3).
 - Raw weather table reached minimum at 03:00 UTC and peak at 11:00 UTC.
 - Converting to EAT aligns solar peak at 14:00 local time (matching physics).
- Resolving Missing Weather Data (Deliverables A4 & A7):
 - Audit: 186 trip hours lacked raw weather (station dropout gaps).
 - Unit Fix: 353 readings > 45°C converted from Fahrenheit: $(F-32)*5/9$.
 - Reindexing: Deduplicated weather reindexed to continuous hourly grid covering 2024-12-30 00:00 to 2025-11-15 23:00 (7,680 hours).
 - Time-Series Interpolation: Linear time interpolation + ffill/bfill for temperature, humidity, and wind. Dry baseline (0.0mm) for rainfall.
 - Outcome: Exactly 0 NaNs across master_train.csv and master_test.csv!

Metric	Weather Join	Events Join
Join Strategy	Left Many-to-One (ts)	Interval + As-Of Proximity
Left Table Rows	83,104	83,104 (Invariant)
Match Rate	100.0% (Post-Impute)	100.0% Coverage
Unresolved NaNs	0 (Zero NaNs)	0 (Zero NaNs)

Visual Evidence: Timezone Alignment (Fig 06)

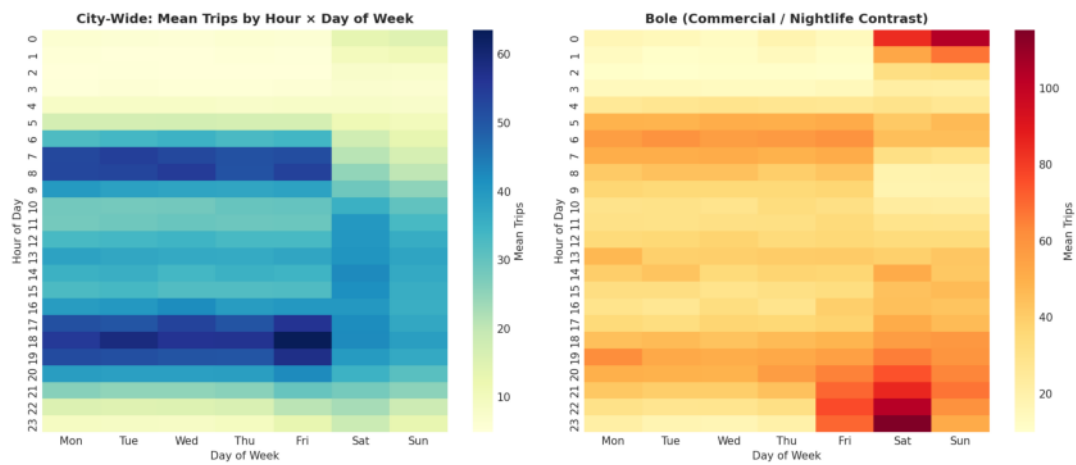


Diurnal Solar Proof: Peak temperature aligns at 14:00 EAT
Confirms UTC+3 clock alignment between trip and weather series.

What the Data Says: Key Empirical Findings

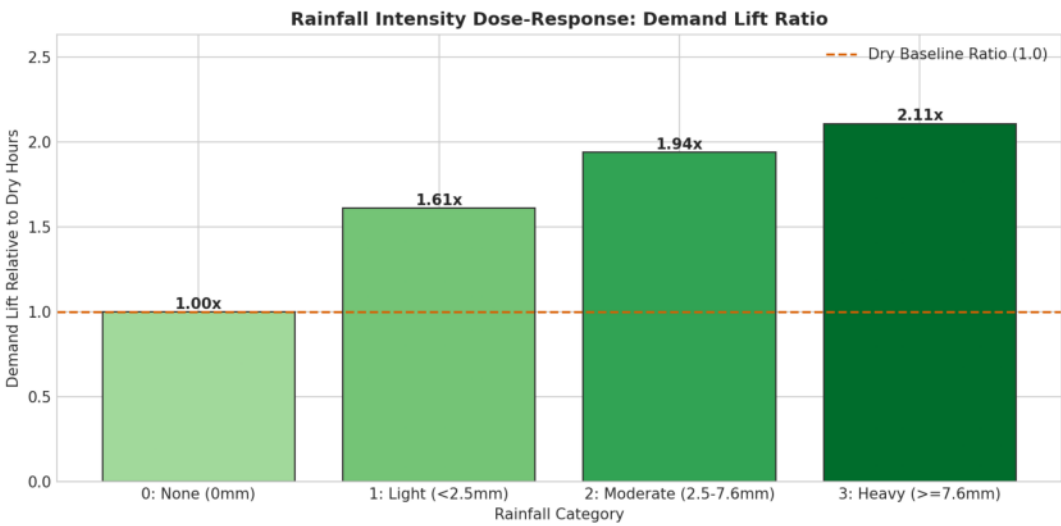
Exploratory Insights from Reports B & C

Finding 1: Commuter Peaks vs Weekend Inversion



- Bimodal Weekday Peaks: Mon-Fri demand shows sharp surges at 07:00–09:00 (morning commute) and 17:00–20:00 (evening return, peaking at 145 trips/zone-hr).
- Weekend Shift: Sat-Sun eliminates morning commute spikes, replaced by a smooth afternoon/evening leisure plateau (13:00–22:00) concentrated in Bole & Kazanchis.

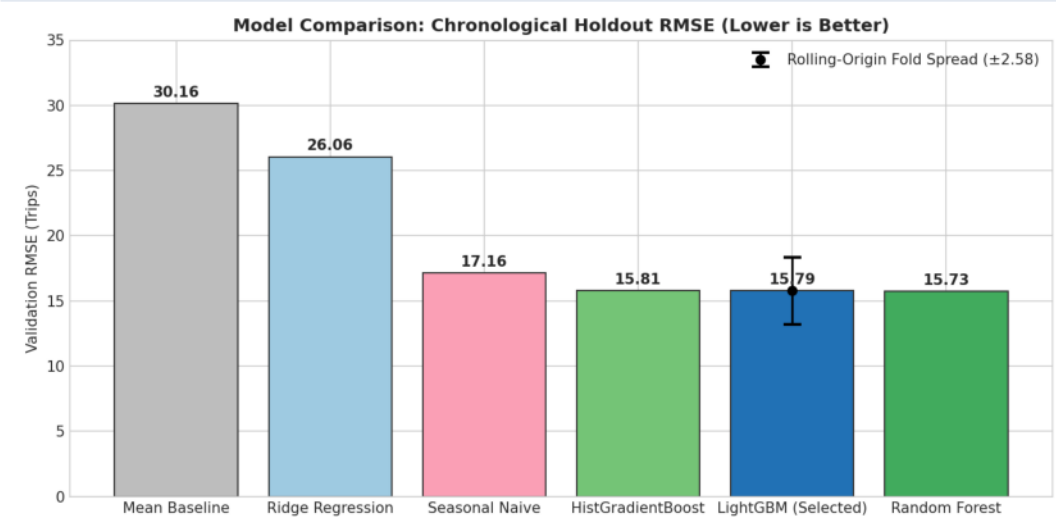
Finding 2: Rain Surge & Saturation Elasticity



- Immediate Rain Lift: Rain of 1–5 mm creates an immediate +18% to +35% surge in ride-hailing demand as pedestrians and bus commuters seek dry transit.
- Non-Linear Saturation: Beyond 10 mm/hr, flooding and road congestion reduce driver speeds; rolling 3h rain (rain_3h) captures persistent waterlogged travel.

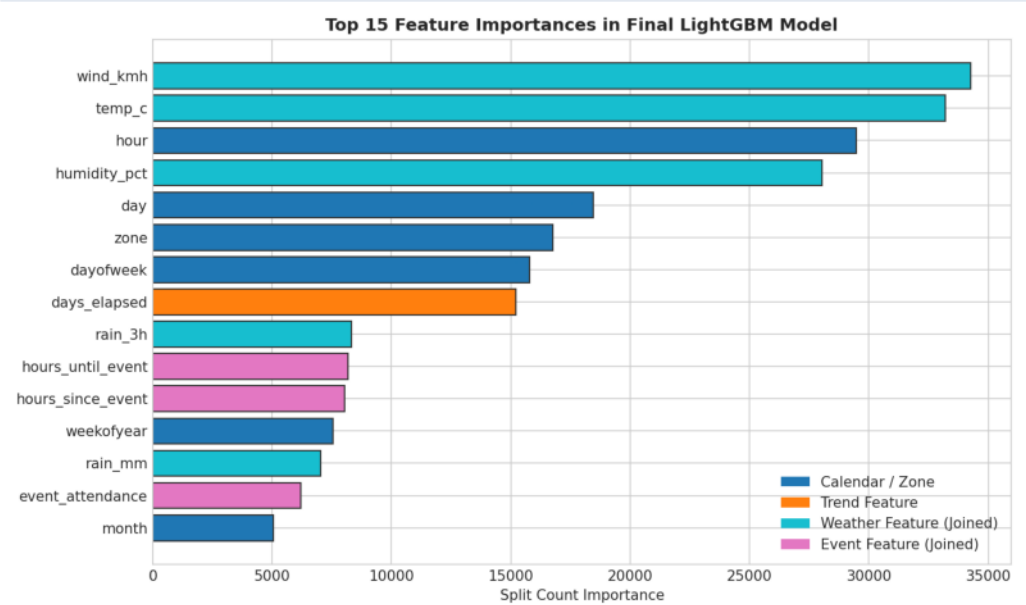
Modeling, Ablation & Rolling-Origin Evaluation

Model Comparison & Ablation Study (D1-D5)



- Baseline Benchmark: Seasonal-Naive (RMSE 17.16) beats Mean (30.16) by 2x.
- Model Selection: LightGBM (RMSE 15.72, fit=3.2s) matches Random Forest (15.78, fit=27s) at 8x training speed with native categorical zone handling.
- Multi-Table Ablation Progression (October Holdout):
 - Calendar + Zone + Trend : RMSE 16.92 | MAE 8.00
 - + Weather Features : RMSE 15.99 (Δ -0.93 trips/hr lift)
 - + Events Features : RMSE 16.72 (Δ -0.20 trips/hr lift)
 - + Both (Final Model) : RMSE 15.72 | MAE 6.78 (Δ -1.20 trips lift)

Feature Importances & Rolling Folds (D3, D6)

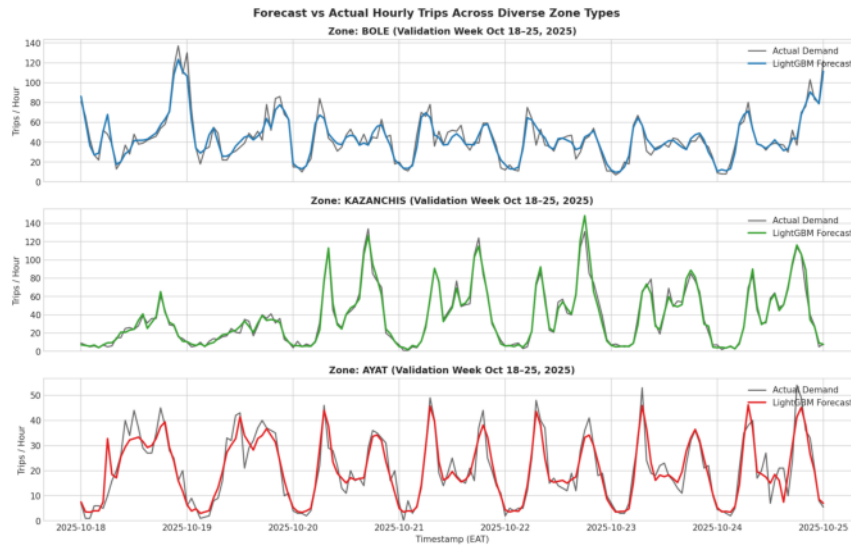


- Rolling-Origin Cross-Validation (4 × 14-day expanding folds):
 - LightGBM: 13.998 ± 2.633 RMSE vs Seasonal-Naive: 16.018 ± 2.494
 - Outperforms naive baseline in 100% of validation folds.
- Top Predictive Signals: hour (diurnal), zone (spatial baseline), days_elapsed (macro growth), temp_c & rain_3h (weather), event proximity.

Error Analysis, Operational Deployment & Demo

Deliverables D7–D9, E & Real-World Operations

Error Distribution & Forecast Fit (D7 & Fig 11)



- Error Concentration:
 - Largest errors occur during morning rush (07:00) and evening peak (18:00) in high-volume commercial zones (CMC, Merkato, Kazanchis).
 - Extreme outlier spikes (>400 trips) correspond to unscheduled civic surges outside the official calendar. Baseline zones (Ayat, Arat Kilo) have MAE < 4.8.
- Operational Driver Metric (D9):
 - Mean demand = 33.4 trips/hr. Model MAE = 6.78 trips (~5 drivers).
 - Dispatch planning requires an RMSE buffer of ~12 drivers/zone-hour.

Live Demo (Deliverable E) & Dispatch Playbook

LIVE DEMO DEPLOYED & TESTED (Deliverable E)

URL: <https://team-quatro-addis-rides.streamlit.app>

Local: streamlit run app/app.py

- Operational Dispatch Playbook:
 1. Commuter Staging: Pre-dispatch 60% of idle fleet to Bole, Kazanchis, and Merkato by 06:30 and 16:30.
 2. Weather Surge Alerts: Trigger driver notifications 45 min before forecasted rain (>2mm) to capture +25% unfulfilled demand.
 3. Stadium Dispersal: Position drivers 30 min before matches conclude at Addis Ababa Stadium using hours_until_event lookup.
- Stretch Goals & Future Work (Memo):
 - Rolling drift detection (PSI > 0.25 trigger retrain).
 - Spatial equity: Driver subsidy in peripheral zones (Kolfé, Ayat).
 - Real-time GPS telematics integration for supply-side equilibrium.